

Solving Trigonometric Equations

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Standard: CCSS.MATH.CONTENT.HSF-TF.B.7 — Use inverse functions to solve trigonometric equations that arise in modeling contexts; evaluate the solutions using technology, and interpret them in terms of the context.



Mr. Kenny's Key Point

Two big traps in trig equations:

1. **Always check the range** (,) — extraneous roots like x must be rejected.
2. For **multiple-angle** equations (x , x , etc.), find ALL solutions in the wider domain first, then translate back to x .

Part A — Direct Equations & Factoring

Directions: Solve each equation on the interval $[0, 2\pi)$ unless stated otherwise. Show all work in the workspace provided.

1. Solve $\sin x = \frac{1}{2}$ on $[0, 2\pi)$.

2. Solve $\sin x = \frac{1}{2}$ on $[0, \pi]$.

3. Solve $\sin \theta = \cos \theta$ on $[0, \pi]$.

Hint: Factor out a common trig factor first.

4. Solve $\sin x = \cos x$ on $[0, \pi]$.

Hint: Divide both sides by $\cos x$ or use the Pythagorean identity.

Part B — Quadratic in a Trig Function

Directions: Let u = the trig function, factor, then solve. Watch for extraneous roots!

5. Solve $\sin^2 x - \sin x = 0$ on $[0, \pi]$.

Let $u = \sin x$. Factor and reject any root outside $[0, \pi]$.

6. Solve $\cos^2 \theta - \cos \theta = 0$ on $[0, \pi]$.

Use the identity $\sin^2 \theta = 1 - \cos^2 \theta$ to convert to a quadratic in $\cos \theta$.

7. Solve x on π .

Part C — Multiple-Angle Equations

Directions: For multiple-angle equations, first solve for the compound angle over an extended domain, then convert back to x .

8. Solve $\cos 2x = \frac{1}{2}$ on $[0, \pi]$.

Let $u = 2x$. Solve for u in $[0, 2\pi]$, then halve.

9. Solve $\sin\left(\frac{x}{2}\right) = \frac{1}{2}$ on $[0, \pi]$.

Isolate $\frac{x}{2}$, solve for x in $[0, \pi]$, then double.

10. Solve $x^4 - 16 = 0$ on $[-\pi, \pi]$.

Let $u = x^2$. Solve for u in $[0, \pi]$, then halve. Expect four solutions.
